

Deutsche Akkreditierungsstelle

Annex to the Partial Accreditation Certificate D-PL-15209-01-01 according to DIN EN ISO/IEC 17025:2018

Valid from: 18.10.2024 **Valid to:** 17.10.2029
Date of issue: 18.10.2024

This annex is a part of the accreditation certificate D-PL-15209-01-00.

Holder of partial accreditation certificate:

Ghana Standards Authority
Shiashie-Legon-Madina Road, Near Gulf House,
P. O. BOX MB 245, ACCRA
GHANA

with the location

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The testing laboratory meets the requirements of DIN EN ISO/IEC 17025:2018 to carry out the conformity assessment activities listed in this annex. The testing laboratory meets additional legal and normative requirements, if applicable, including those in relevant sectoral schemes, provided that these are explicitly confirmed below.

The management system requirements of DIN EN ISO/IEC 17025 are written in the language relevant to the operations of testing laboratories and they conform to the principles of DIN EN ISO 9001.

This certificate annex is only valid together with the written accreditation certificate and reflects the status as indicated by the date of issue. The current status of any given scope of accreditation can be found in the directory of accredited bodies maintained by Deutsche Akkreditierungsstelle GmbH at <https://www.dakks.de>.

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Tests in the fields:

**physical, physico-chemical, chemical and microbiological analysis of food, animal feeding stuff and environmental samples in the area of food production and food handling;
selected physical, physico-chemical, chemical analysis of cosmetics**

The testing laboratory is permitted, without being required to inform and obtain prior approval from DAkkS, to use standards or equivalent testing methods listed here with different issue dates.

The testing laboratory maintains a current list of all testing methods within the flexible scope of accreditation.

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1 Physical, physico-chemical, chemical analysis of food

1.1 Determination of pesticide residues in vegetable oils and Low-Fatty Matrices

GSA-SM-T03-E 2021-05	Multiresidue Determination of Pesticides in Low-Fatty Matrices with GC-ECD and GC-PFPD
GSA-SM-T04-D 2022-10	Determination of pesticide residues in vegetable oils by LC-MS/MS
GSA-SM-T14-D 2022-10	Determination of pesticide residues in fruits and vegetables (both with high water content) by LC-MS/MS

1.2 Determination of Aflatoxins in cereals, nuts and derived products, milk and milk products using high-performance liquid chromatography by conventional fluorescence detector

ISO 16050 2003-09	Foodstuffs - Determination of aflatoxin B1, and the total content of aflatoxins B1, B2, G1 and G2 in cereals, nuts and derived products - High-performance liquid chromatographic method (Modification: <i>Extraction solvent ratio 80/20, different derivatization by Kobra cell and Potassium Bromide</i>)
GSA-SM-T131-A 2019-10	Determination of Aflatoxin M1 in milk and milk products using high-performance liquid chromatography by fluorescence detector

1.3 Determination of Histamine and biogenic amines in fish and fishery products using high performance liquid chromatography by conventional fluorescence-detector

GSA-SM-T21-D 2018-07	Method for the determination of Histamine in fish and fishery products using High Performance Liquid Chromatography by conventional Fluorescence-Detector (Xenon lamp)
GSA-SM-T80-A 2019-01	Method for determination of Biogenic amines in fish and fishery products using High Performance Liquid Chromatography by conventional Fluorescence-Detector (Xenon lamp)

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1.4 Determination of Mycotoxins in juices and their concentrates using high performance liquid chromatography by conventional UV-detector

ISO 8128-1
1993-07

Apple juice, apple juice concentrates and drinks containing apple juice - Determination of Patulin content - Part 1: Method using high-performance liquid chromatography
(Modification: *using acetonitrile as extraction solvent; AFLAPAT MYCOSEP SPE Column for purification; 50µl injection volume; 90:10 (Acetonitrile: Water) standard solvent and mobile phase*)

1.5 Determination of elements in fish and fishery products using atomic absorption spectrometry (AAS)

BS EN 13806
2002-09

Foodstuffs. Determination of trace elements. Determination of mercury by cold-vapour atomic absorption spectrometry (CVAAS) after pressure digestion
(Limitation: *here only for fish and fishery products*)

1.6 Determination of elements in food using inductively coupled plasma mass spectrometry (ICP-MS)

BS EN 15763
2009-12

Foodstuffs - Determination of trace elements -Determination of lead, cadmium, Arsenic and Mercury in Fish and Fishery Products by Inductively Coupled Plasma Mass spectrometry (ICP-MS) after microwave digestion
(Limitation: *Fish and Fishery Products, Cereals and grains, fruit juice and fruit purees; Mercury is limited to fish and fishery products*)

1.7 Determination of acidity, soluble solids, moisture contents and ash in non-alcoholic beverage, cereals and derived products using physical, physico-chemical testing methods

ISO 712
2009-11

Cereals and cereal products - Determination of moisture content - Reference method

ISO 750
1998-08

Fruit and vegetable products - Determination of titratable acidity
(Modification: *here for Non-alcoholic beverage*)

ISO 3960
2017-02

Determination of peroxide value – iodometric (Visual) endpoint determination in animal and vegetable fats and oils and fats and oils products (Modification: *uses chloroform instead of iso-octane*)

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AOAC Method 923.03 2019	Ash of Flour (Modification: <i>here for Non-alcoholic beverage</i>)
GSA-SM-T54 2022-10	Determination of Crude Protein Content in Food and Agricultural products (Limitation: <i>here only for food</i>)
GSA-SM-T140 2022-10	Determination of Brix (soluble solids) in Fruit Juices an Non-alcoholic beverage using ATAGO PEN-PRO refractometer

2 Cultural-microbiological determination of bacteria in food, animal feeding stuff and environmental samples in the area of food production and food handling

ISO 4831 2006-08	Microbiology of food and animal feeding stuffs - Horizontal method for the detection and enumeration of coliforms - Most probable number technique (Modification: <i>here also for environmental samples in the area of food production and food handling</i>)
ISO 4832 2006-02	Microbiology of food and animal feeding stuffs - Horizontal method for the enumeration of coliforms - Colony-count technique
ISO 4833-1 2013-09	Microbiology of the food chain - Horizontal method for the enumeration of microorganisms - Part 1: Colony count at 30 degrees C by the pour plate technique (Modification: <i>here also for environmental samples in the area of food production and food handling</i>)
ISO 4833-2 2013-09	Microbiology of the food chain - Horizontal method for the enumeration of microorganisms - Part 2: Colony count at 30 degrees C by the surface plating technique
ISO 6579-1 2017-02	Microbiology of the food chain - Horizontal method for the detection, enumeration and serotyping of Salmonella - Part 1: Detection of Salmonella spp. (Modification: <i>here also for environmental samples in the area of food production and food handling</i>) (Limitation: <i>without Annex D</i>)
ISO 6888-2 2021-08	Microbiology of the food chain — Horizontal method for the enumeration of coagulase-positive staphylococci (Staphylococcus aureus and other species) Part 2: Method using rabbit plasma fibrinogen agar medium (Limitation: <i>only for food</i>)

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ISO 7251 2005-02	Microbiology of food and animal feeding stuffs - Horizontal method for the detection and enumeration of presumptive <i>Escherichia coli</i> - Most probable number technique
ISO 16649-2 2001-04	Microbiology of food and animal feeding stuffs - Horizontal method for the enumeration of beta-glucuronidase-positive <i>Escherichia coli</i> - Part 2: Colony-count technique at 44 degrees C using 5-bromo-4-chloro-3-indolyl beta-D-glucuronide (Limitation: <i>only for food</i>)
ISO 21527-1 2008-07	Microbiology of food and animal feeding stuffs - Horizontal method for the enumeration of yeasts and moulds - Part 1: Colony count technique in products with water activity greater than 0,95 (Limitation: <i>only for food</i>)
ISO 21527-2 2008-07	Microbiology of food and animal feeding stuffs - Horizontal method for the enumeration of yeasts and moulds - Part 2: Colony count technique in products with water activity less than or equal to 0,95 (Limitation: <i>only for food</i>)
ISO 21528-2 2017-06	Microbiology of the food chain - Horizontal method for the detection and enumeration of Enterobacteriaceae - Part 2: Colony Count technique (Modification: <i>here also for animal feeding stuff</i>)

3 Selected Physical and physico-chemical analysis of cosmetics

ISO 456 1973-03	Surface active agents – Analysis of soaps – Determination of free caustic alkali by Titration
ISO 685 2020-02	Analysis of soaps – Determination of total alkali content and total fatty matter content by Gravimetric method (Limitation: <i>only for total fatty matter content</i>)

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Abbreviations used:

AOAC	Association of Official Agricultural Chemists
BS	British Standards Institution
DIN	Deutsches Institut für Normung e.V. - German institute for standardization
EN	European Union
IEC	International Electrotechnical Commission
ISO	International Organisation for Standardization
GSA-SM-XXX-X	House procedures of the Ghana Standards Authority Biochemical Science Laboratory